Continual Learning for Adaptive Health and Reuse Assessment of Heterogeneous Second-Life EV Batteries

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***Abstract*—Retired electric-vehicle batteries can serve in second-life applications, but a health model trained on one battery population may fail on the next, and continual learning promises to adapt without forgetting. We test this promise for forecasting the state of health (SOH) of lithium-ion cells one to six capacity checks ahead, on a leakage-controlled benchmark built from five public datasets (395 cells after quality rules), with two label artefacts removed, every follow-up experiment pre-registered, and every claimed difference required to replicate on a second seed set, after a re-run showed that several significance results depended on the software environment. A Temporal Fusion Transformer with Elastic Weight Consolidation (TFT + EWC) is compared with fine-tuning, replay, joint training, literature backbones and simple forecasters on a continual stream, on unseen usage conditions and after first-life pretraining. Forgetting appears whenever usage conditions differ (0.18–0.46 percentage points, pp). Trained on a shrunk correction to linear extrapolation of the recent fade rate, TFT + EWC matches extrapolation on the stream, where no model beats it, and beats it by 0.12–0.20 pp on unseen conditions; there, however, gradient boosting fitted on all conditions is 0.39–0.60 pp more accurate, and the same TFT trained jointly beats its continual form by 0.32–0.44 pp. Shift-adaptive EWC, first-life pretraining and inputs from 0.9 billion ageing measurements processed with Apache Spark do not help. Conformal forecast intervals calibrated on the training conditions under-cover on unseen ones (0.66–0.76 at 90%); calibrating on conditions held out from fitting restores 0.90 coverage over the unseen conditions, but not on the one outside the training range (0.38–0.48). With these intervals, 87–92% of forecasts fall within one reuse class, and over 99% of those classes are correct.**

***Index Terms*—Continual learning, second-life batteries, electric vehicles, state of health, battery degradation, heterogeneous batteries, catastrophic forgetting, machine learning**

I. Introduction

Lithium-ion batteries lose capacity and power with cycling, temperature and time. When an electric-vehicle (EV) battery no longer meets automotive requirements, it may still serve less demanding applications such as stationary storage, renewable-energy buffering or backup power. Reuse extends the battery's life and delays recycling, but only if its health is assessed reliably: a battery that suits one application may not suit another.

The state of health (SOH) describes the present capability of a battery relative to a reference condition, and machine-learning models have estimated it from voltage, current, temperature, partial charging and impedance measurements [1]–[4]. Retired batteries are, however, heterogeneous: chemistry, manufacturer, capacity, first-life history, temperature and charging protocol differ, and complete first-life records are often missing. A model trained once on a fixed population may therefore fail on the next population. Continual learning addresses this by updating a model on new data while preserving what it learned earlier; it must therefore be evaluated both on the newly introduced battery group and on the groups learned before, to measure catastrophic forgetting.

This work combines continual learning, health assessment and second-life reuse assessment. Health is assessed by *forecasting* the SOH one to six capacity checks ahead from everything measured up to the latest check: on the first dataset studied, estimating the current SOH from cycling features was beaten by carrying the last measured SOH forward (§III), whereas a reuse decision needs the future SOH. The forecast is mapped to reuse-suitability classes.

The contributions are:

- A leakage-controlled benchmark for forecasting the SOH of second-life cells, built from five public datasets (395 cells after quality rules) verified against their repository records, with automated data checks and protocols fixed before model training.
- Two label artefacts identified and removed: test currents set from the previous capacity, which encode the label, and pulse charge counted in the capacity, which shifts labels by up to 0.09 percentage points.
- An evaluation of a Temporal Fusion Transformer with Elastic Weight Consolidation (TFT + EWC) against fine-tuning, replay, joint training, simple forecasters and the model families of the literature (BiLSTM, TCN, Trans-

former, GPR, random forest, SVR) under three protocols (a continual stream, unseen groups and first-life pretraining), reporting adaptation, forgetting and retention.
- A continual TFT + EWC that learns a shrunk correction to linear extrapolation: on the continual stream it matches extrapolation, which no model beats, and EWC reduces its forgetting over longer horizons; on unseen usage conditions it beats extrapolation but not gradient boosting or its own jointly trained form (all replicated). Shift-adaptive EWC and first-life pretraining add nothing, null results we report.
- Forecast intervals for unseen usage conditions: split-conformal intervals calibrated on the training conditions under-cover, and calibrating on conditions held out from fitting restores nominal coverage over the unseen conditions, except for a condition outside the training range.
- A reproducibility analysis showing that several significance results on small battery datasets depend on the numerical environment, and a replication rule that the main results satisfy.

II. Related Work

The review covers 2022–2026 studies on second-life lithium-ion batteries, SOH estimation, continual learning and reuse assessment (Table I).

SOH estimation for second life. Braco *et al.* estimated the SOH of second-life cells from partial charging under residential and fast-charging-station profiles, with offline RMSE of about 0.6 and 0.8% and 85% less testing time than a full capacity test [1]. In a companion study on six reused Nissan Leaf modules they compared 58 health indicators from incremental-capacity analysis, partial charging, constant-current/constant-voltage stages and internal resistance; the best indicator at repurposing reached about 0.2% RMSE, but the most useful indicators changed as ageing progressed [2]. Faraji-Niri *et al.* accelerated second-life SOH estimation with electrochemical impedance spectroscopy (EIS) and machine learning [3]; EIS is fast and non-destructive but sensitive to temperature, state of charge and equipment. Takahashi *et al.* screened batteries for second life with statistical current and voltage features and bagged Gaussian process regression, with root-mean-squared percent error below 1.48% in the worst cases [4]. Cui *et al.* tested retired Nissan Leaf cells for more than 15 months under grid-storage-like operation and estimated health online with clustering, with mean absolute percentage error below 2.3% [5]. Vignesh *et al.* reviewed SOH estimation methods for second-life batteries and their challenges [6]. These studies estimate the *current* SOH of a fixed population; none adapts a model as new, different populations arrive.

Continual learning and generalisation. Lee *et al.* applied bilevel-optimised continual learning to predict the capacity degradation of lithium-ion batteries [7]. Yao *et al.* used progressive neural networks with temporal convolutional networks for online state-of-charge estimation across three public drive-cycle datasets, explicitly addressing catastrophic forgetting and task order [8]; the target, however, is state of charge, not SOH or reuse. Li *et al.* proposed domain-generalised SOH estimation for second-life EV batteries [9], and Meskine *et al.* compared BiLSTM, TCN and probabilistic TCN-GPR models on sparse, noisy data, where isotonic recalibration substantially reduced the calibration error [10]. Transfer and domain generalisation do not by themselves address a long sequence of tasks with explicit retention.

TABLE I
Summary of Reviewed Literature

Study	Methodology	Main Challenge
Braco 2022 [1]	Partial-charge / real-profile SOH	Depends on charging profile
Braco 2022 [2]	Health-indicator selection	Indicators change with ageing
Faraji-Niri 2023 [3]	EIS + machine learning	Measurement conditions
Takahashi 2023 [4]	Features + bagged GPR	Feature/data quality
Cui 2024 [5]	Online ML + clustering	Limited experimental population
Vignesh 2024 [6]	SOH methods review	No single adaptive model
Lee 2024 [7]	Bilevel-optimised CL	Adaptation versus retention
Yao 2025 [8]	PNN + TCN continual learning	Focused on SOC
Li 2025 [9]	Domain generalisation	Not continual learning
Meskine 2026 [10]	BiLSTM/TCN + probabilistic model	Sparse/noisy data

Research gap. Second-life SOH estimation, screening, transfer learning and degradation prediction have progressed largely separately; continual learning is rarely combined with heterogeneous second-life SOH forecasting, an explicit reuse decision and an evaluation of forgetting. This work asks whether a continual learner can learn heterogeneous second-life populations in sequence while retaining earlier ones, and whether its forecasts support reuse decisions.

III. Methodology

Given everything measured up to a reference performance test (RPT, or capacity check), we forecast the SOH one to three checks ahead, and up to six in the follow-up experiments. Cells or usage conditions arrive as a stream of tasks, and a model must adapt to each new task without forgetting the earlier ones. Fig. 1 summarises the framework.

Change of task after inspecting the data. The study was first designed to estimate the *current* SOH from cycle-level features. On the first dataset these features carried almost no health information (median within-cell Spearman correlation with SOH 0.09, against -0.99 for elapsed time), and learned estimators were worse than persistence (1.78 and 2.37 pp for gradient boosting and ridge regression, against 0.99 pp). We therefore reframed the task as forecasting, after seeing the data; the forecasting protocol was then committed before any forecasting model was fitted, and deviations are reported.

A. Datasets

Table II lists the five public datasets (all CC BY 4.0). Every file is verified against its repository record or checksums, and

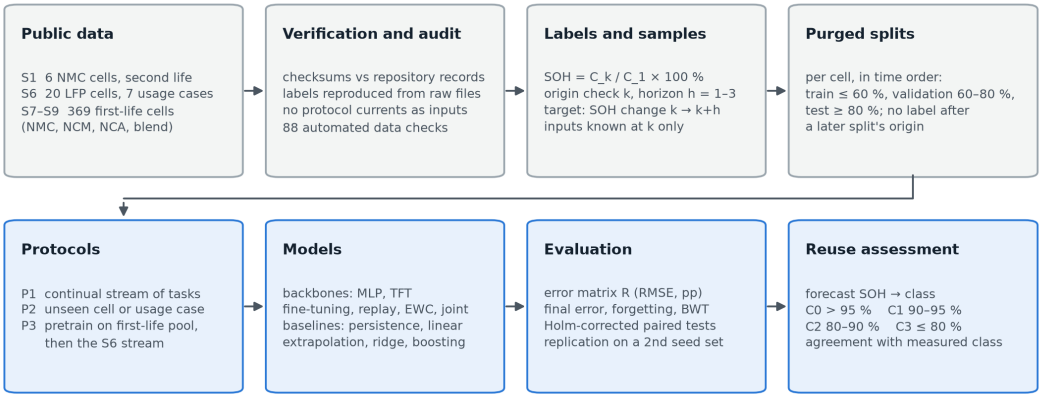


Fig. 1. Framework of the study. Public data are verified and audited, converted into forecasting samples with purged chronological splits, and evaluated under three protocols with continual-learning metrics, a replication rule and a reuse-class mapping.

TABLE II
DATASETS. S1 AND S6 FORM THE CONTINUAL-LEARNING STREAMS; S7–S9 FORM THE FIRST-LIFE PRETRAINING POOL.

Set	Cells and conditions	Label	Samples
S1 [11], [12]	6 NMC 21700 cells, second life on residential or commercial peak shaving, 20/35 °C blocks	C/20 RPT capacity, pulses excluded	279
S6 [13], [14]	20 LFP 18650 cells, 7 conditions (charge mode, voltage and rate, discharge, depth), 50 °C	C/25 check capacity	2,409
S7 [15], [16]	195 of 228 NMC/C+SiO cells, calendar, cyclic and WLTP ageing, 0–40 °C	check-up capacity	10,251
S8 [17], [18]	55 NCM cells, six charge/discharge protocols	capacity cycles	9,699
S9 [19], [20]	119 of 130 NCA, NCM and blended cells, 25–45 °C	capacity cycles	16,761

our parsing of the raw files reproduces exactly the RPT capacities of the processed S1 tables and the authors' capacities of all 843 S6 checks.

S1 (Stanford) contains six LG Chem INR21700-M50T cells that spent 23 months under an EV driving profile with different charge rates [21] and then about 27 months on synthetic peak-shaving duty cycles [22]; each RPT is a C/20 discharge interrupted by C/2 pulses at three voltages, with impedance spectroscopy (EIS) from the fifth RPT [11]. Its heterogeneity lies in first-life ageing and duty cycle among cells of one type. Data-quality findings are handled by fixed rules: RPT 2 (run at about 20 °C) is excluded, off-protocol pulses and galvanostatic spectra are set to missing, and RPT 7's aggregate EIS files are decoded and checked against the authors' values.

S6 (Wheeler *et al.*) was selected by criteria fixed before screening: a reuse licence, a series of capacity checks, ageing past first-life end of life, heterogeneity that S1 lacks and verifiable metadata. Its cells were characterised every 100 cycles and end at 32–64% SOH; "second life" here means continued ageing below 80% SOH under the same protocol. A check is used only if its C/25 discharge starts from full charge and reaches the cut-off (843 of 847 checks).

S7–S9 are first-life datasets used for pretraining. Cells with at least five valid checks are kept: 369 cells and 36,711 samples.

B. Big-data processing of the ageing cycles

Between checks, every S6 cell logged each ageing cycle at about one sample per second: 827 files (one per cell and 100-cycle block), 132 GB raw and 9.0×10^8 rows. They are reduced to block-level inputs with Apache Spark 4.2 (PySpark) in local mode (Fig. 2). In the map stage, one Spark task per block parses the raw rows into steps and derives, for every cycle, a validity flag and seven measured responses: end-of-charge and end-of-discharge voltage, constant-voltage hold time and final current, relaxation over the rest after charge and after discharge, and internal resistance where recorded. Charge, energy and current are excluded because the protocol scales them with the previous capacity. The per-cycle table (about 83,000 cycles) is cached, and Spark SQL then groups the valid cycles by cell and block into the mean, last value and least-squares slope of each response (21 inputs), requiring at least 50 valid cycles. The block ending at a check joins the origin at that check; every block is verified to end before it. On six worker cores, the full run (827 map tasks, 82,529 cycles, 97% valid) takes 20 min. Its output matches an independent pandas implementation on all 827 blocks within 10^{-9} relative plus 10^{-12} absolute (largest difference 2×10^{-13}); a purely relative criterion, stated first, failed only on a slope that is zero up to rounding. Discharged charge, which is excluded, correlates at $r = 0.97$–0.999 within cell with the previous check's capacity, confirming the protocol scaling.

C. State of health and forecasting target

For cell b with check capacities $C_{b,1}, C_{b,2}, \ldots$ in time order,

$$\mathrm{SOH}_{b,k} = 100\, C_{b,k}/C_{b,1}\ \%, \tag{1}$$

i.e. relative to the first valid measurement, not to a nominal rating; for S1 this is already an aged state. In S1 the capacity is

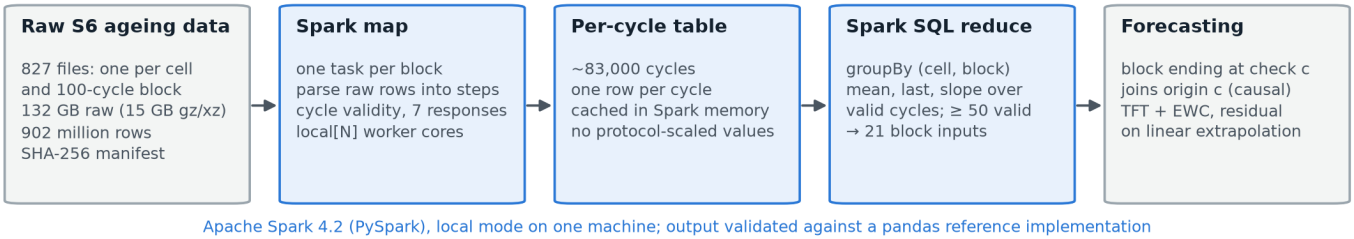


Fig. 2. Big-data pipeline for the S6 ageing cycles: Spark map (one task per 100-cycle block) to a cached per-cycle table, Spark SQL reduction to block inputs, and a causal join to the forecasting origins.

the charge of the C/20 steps only [11, Eq. (1)]. The processed tables we started from also counted the pulse charge, which shifts some labels by up to 0.09 pp, comparable to one RPT's fade.

A sample is a triple (cell b, origin k, horizon $h \in \{1, 2, 3\}$), about one to three months in S1 and 100–300 cycles in S6. The target is

$$\Delta_{b,k,h} = \mathrm{SOH}_{b,k+h} - \mathrm{SOH}_{b,k}, \tag{2}$$

so an error in Δ equals the error in the future SOH; errors are in SOH percentage points (pp). Over a set $\mathcal{D}$ of N samples,

$$\mathrm{RMSE}(\mathcal{D}) = \sqrt{\tfrac{1}{N} \textstyle\sum_{\mathcal{D}} \left(\hat{\Delta}_{b,k,h} - \Delta_{b,k,h}\right)^2}. \tag{3}$$

The fade rate $r_{b,k}$ is the least-squares slope of SOH against time t (days or cycles) over the last three valid checks,

$$r_{b,k} = \frac{\sum_{i=k-2}^{k} (t_{b,i} - \bar{t})(\mathrm{SOH}_{b,i} - \overline{\mathrm{SOH}})}{\sum_{i=k-2}^{k} (t_{b,i} - \bar{t})^2}. \tag{4}$$

Inputs use only information available at the origin. **F0** contains the SOH history ($\mathrm{SOH}_{b,k}$, time or cycles since the first check, $r_{b,k}$), static descriptors (S1: duty cycle and first-life history [11]; S6: test conditions) and the *planned* usage until the target (S1: days at each temperature set point and at rest; S6: cycles), so forecasts are conditional on the planned usage. **F1** adds diagnostics measured at the origin: 47 in S1 (pulse resistances, charge between pulse voltages, incremental-capacity peaks of the C/20 discharge and EIS features) and 17 in S6 (plateau charges, incremental-capacity peaks, hysteresis, a resistance proxy and capacity ratios). No test current is an input: in S1 the RPT and monthly discharge currents were set from the previous capacity and therefore encode the label. Missing values are imputed from training data and flagged.

D. Splits and protocols

Purged chronological splits. For each cell, cut A lies at 60% and cut B at 80% of its valid checks. Training samples have their target at or before A, validation samples their origin at or after A and target at or before B, and test samples their origin at or after B; samples straddling a cut are dropped, so no training label lies after a validation origin and no validation label after a test origin.

P1, continual stream. A task is a cell in S1 (six tasks) or a test condition in S6 (seven tasks). At step i of an ordering π, a strategy receives only task π_i's data. Each method runs on 10 orders with 5 seeds, identical across methods.

P2, unseen group. One cell (S1) or condition (S6) is held out; models are trained on the others and tested on all its samples.

P3, first-life pretraining. All datasets are mapped onto shared inputs: SOH history, calendar and cycle time, planned time to the target and operating conditions (temperature, charge and discharge rates, depth of discharge, charge voltage and CV phase, storage SoC, chemistry), with undefined conditions flagged as missing. Models are pretrained on the pool, which also supplies the input scaling and the first EWC anchor; the seven S6 conditions then follow as a stream. We compare training from scratch, pretraining with fine-tuning and pretraining with EWC for both backbones, and record the pool test error before and after the stream.

E. Models

Preprocessing (standardisation, median imputation with missing indicators, target scaling) is fitted on the first task and then frozen. The *MLP* has hidden layers of 128 and 64 ReLU units (dropout 0.1). The proposed backbone adapts the Temporal Fusion Transformer (TFT) [23]: a window of the $W{=}4$ most recent origins of the same cell and horizon (inputs only, never labels) passes through per-variable embeddings, variable selection by gated residual networks, an LSTM encoder [24] and interpretable two-head attention masked to past steps.

Strategies. Fine-tuning trains on the current task only (lower bound); experience replay [25] stores B samples per finished task; joint training fits all tasks at once (upper bound); and Elastic Weight Consolidation (EWC) [26] adds, for each finished task j,

$$\frac{\lambda}{2} \sum_{p} F_{j,p} \left(\theta_p - \theta^*_{j,p}\right)^2, \tag{5}$$

with θ^*_j the parameters after task j and F_j the empirical diagonal Fisher information over up to 500 training samples. *TFT + EWC* is the proposed model. Training uses Adam [27], batches of 64, at most 200 epochs and early stopping on the current task's validation RMSE.

Reference forecasters are persistence ($\hat{\Delta} = 0$), linear extrapolation $\hat{\Delta}_{b,k,h} = r_{b,k}(t_{b,k+h} - t_{b,k})$, ridge regression (on planned usage only, F0 or F1) and histogram gradient boosting on F1.

Learned residual. Because linear extrapolation proved hardest to beat, a follow-up experiment, specified and committed before it was run, trains the continual learners on the correction to it. With $\hat{\Delta}^{\text{lin}}_{b,k,h}$ the extrapolation above (zero when $r_{b,k}$ is missing), the model g_θ learns $\Delta_{b,k,h} - \hat{\Delta}^{\text{lin}}_{b,k,h}$ and the forecast is

$$\hat{\Delta}_{b,k,h} = \hat{\Delta}^{\text{lin}}_{b,k,h} + g_\theta(\mathbf{x}_{b,k,h}). \tag{6}$$

Since $\hat{\Delta}^{\text{lin}}$ is known at the origin, every error is unchanged by this reparameterisation. The residual and the direct (target Δ) forms are compared on the S6 stream with the P3 shared inputs, from scratch, for horizons 1–3 (E1) and 1–6 (E2), each tuned and evaluated as P3.

Shift-adaptive EWC and literature backbones. A second pre-registered experiment tests whether the EWC strength should follow the shift between groups. *RC-TFT* trains the residual TFT with shift-adaptive EWC (SA-EWC): when task t starts, the penalty (5) of each finished task j is multiplied by $d_{j,t} = \big((\|\mu_t - \mu_j\|^2 + \|\sigma_t - \sigma_j\|^2)/D\big)^{1/2}$, the per-dimension 2-Wasserstein distance between diagonal Gaussians fitted to the two tasks' preprocessed training inputs, with λ_0 tuned as λ. Alongside, BiLSTM [10], TCN [8], [10] and Transformer-encoder continual learners (fine-tuning and EWC, the TFT's input window, within 20% of its 12,708 parameters) and single-fit GPR [3], [4] (at most 1,000 rows), random forest and SVR regressors are run in residual and direct form. SA-EWC counts as positive only if it improves on fixed-λ EWC in final error or forgetting, replicated, without a replicated loss on the other.

Shrinkage and forecast intervals. A third pre-registered experiment calibrates the residual model after training, without changing its weights. Per horizon h, the learned correction is shrunk towards extrapolation by $\alpha_h = \min\big(1, \max(0, \sum p\,y / \sum p^2)\big)$, with p the predicted and y the observed correction on calibration rows. A split-conformal interval [28] $\hat{\Delta} \pm q_h$ takes as q_h the $\lceil (n+1)(1-a) \rceil$-th smallest of the n absolute calibration residuals at horizon h. Intervals at level $1-a$ (90% and 80%) are scored by coverage, width and the interval score [29] $2q_h + \frac{2}{a}\max(|\Delta - \hat{\Delta}| - q_h, 0)$ (pp; lower is better). On the stream, both are refitted after each task on its validation rows, and extrapolation gets intervals in the same way. The same experiment pretrains the residual model on the first-life pool.

Unseen conditions. A fourth experiment repeats P2 on S6 for the shrunk residual TFT + EWC: it is streamed over the six training conditions in the ten evaluation orders with the P1 tuned values, and shrinkage and intervals are fitted once on the pooled validation rows of those conditions. Since extrapolation equals persistence at a cell's first two checks, where no fade rate exists, origins with a fade rate (H^+) and without (H^-) are reported separately. A fifth experiment calibrates intervals by leaving one condition out: each training condition is predicted by a model fitted on the other five, and the absolute residuals at all its origins, pooled over the six conditions, give q_h, a cross-conformal variant [30], optionally per horizon and stratum (Mondrian). Its forecaster is gradient boosting, the most accurate in P2; being deterministic, it must show a result in both horizon sets instead of two seed sets.

F. Hyperparameters, metrics and statistics

Hyperparameters are chosen on validation data from tuning orders disjoint from the evaluation orders: the learning rate per backbone (lowest validation plasticity $\frac{1}{T}\sum_j R_{j,j}$ of fine-tuning streams), then replay $B \in \{50, \dots, 400\}$ and EWC $\lambda \in \{10^0, \dots, 10^5\}$; a selection at a grid edge extends the grid. For P3, ten tuning orders are used with the one-standard-error rule

$$g^\star = \min\big\{\, g : \bar{m}_g \le \bar{m}_{\hat{g}} + s_{\hat{g}} \big\}, \quad \hat{g} = \arg\min_g \bar{m}_g, \tag{7}$$

where $\bar{m}_g$ is the mean validation criterion of grid value g and s_g its standard error.

After each step i, every task's test set is evaluated, giving $R_{i,j} = \text{RMSE}(\mathcal{D}^{\text{te}}_{\pi_j} \mid \theta_i)$. Following [31], [32], we report the final average error $\frac{1}{T}\sum_j R_{T,j}$, backward transfer $\frac{1}{T-1}\sum_{j<T}(R_{T,j} - R_{j,j})$ and forgetting $\frac{1}{T-1}\sum_{j<T}(R_{T,j} - \min_{j \le i < T} R_{i,j})$ (positive values mean forgetting), the skill $1 - \text{RMSE}/\text{RMSE}_{\text{persistence}}$, and the agreement of the reuse class of the forecast SOH with the measured class (C0 $> 95\%$, C1 90–95%, C2 80–90%, C3 $\le 80\%$, fixed in advance).

Means carry 95% bootstrap intervals [33]. Methods are compared by two-sided Wilcoxon signed-rank tests [34] on matched runs or held-out groups, with rank-biserial effect sizes [35] and Holm correction [36]. A difference d is *replicated* only if it is significant in the main seed set (0–4) and in a replicate set (5–9), with the same sign and $|d| \ge 0.05$ pp in both; $|d| < 0.05$ pp in both means practically equivalent. The threshold is about half the label shift caused by the pulse-charge definition.

G. Implementation and availability

The pipeline uses Python 3.11, NumPy 2.4, pandas 3.0, scikit-learn 1.9 and PyTorch. P1 and P2 were first run on an Intel Xeon (PyTorch version not recorded) and re-run, like P3 and the ablation, with PyTorch 2.14.1 on an Intel Core 5 210H, with one CPU thread and seeded, deterministic algorithms. Every result file records configuration, seeds, library versions and the Git commit. Unit tests cover input causality, split purging and the exclusion of labels and protocol currents from the inputs. The datasets are at doi:10.17605/OSF.IO/8JNR5 (S1), 10.57745/OLBXKT (S6), 10.35097/1969 (S7), 10.5281/zenodo.10963339 (S8) and 10.5281/zenodo.6405084 (S9); code, decision log and results are kept in a private repository and are available from the authors on request.

IV. Results

Data. S1 diagnostics track the current SOH (median within-cell $\rho = 0.97$ for the main incremental-capacity peak) but not the change *to the next RPT* (median $|\rho| \le 0.3$), and every S1 sample stays in class C0. S6 conditions fade very differently, and all cells cross from C0 to C3.

TABLE III

PROTOCOL P1 (CONTINUAL STREAM). FINAL AVERAGE TEST RMSE OF SOH($k+h$) IN PP (MEAN ± S.D. OVER 10 ORDERS × 5 SEEDS FOR CONTINUAL METHODS, OVER 5 SEEDS FOR JOINT TRAINING), SKILL RELATIVE TO PERSISTENCE ($1 - \mathrm{RMSE}/\mathrm{RMSE}_{\mathrm{persistence}}$) AND FORGETTING (PP). †UPPER BOUND, TRAINED ON ALL TASKS AT ONCE.

	S1: six NMC second-life cells			S6: 20 LFP cells, seven usage conditions		
Method	RMSE	Skill	Forgetting	RMSE	Skill	Forgetting
Persistence	0.632	0.00	–	1.644	0.00	–
Linear extrapolation	0.726	-0.15	–	0.343	0.79	–
Ridge, usage only	0.450	0.29	–	1.274	0.22	–
Ridge, F0	0.678	-0.07	–	0.801	0.51	–
Ridge, F1	0.491	0.22	–	0.605	0.63	–
Gradient boosting, F1	0.466	0.26	–	0.372	0.77	–
MLP fine-tuning	0.492 ± 0.029	0.22	0.030	0.531 ± 0.151	0.68	0.207
MLP replay	0.475 ± 0.021	0.25	0.025	0.597 ± 0.178	0.64	0.212
MLP + EWC	0.497 ± 0.033	0.21	0.040	0.540 ± 0.176	0.67	0.212
MLP joint†	0.500 ± 0.016	0.21	–	0.414 ± 0.037	0.75	–
TFT fine-tuning	0.467 ± 0.014	0.26	0.026	0.883 ± 0.538	0.46	0.461
TFT + EWC	0.462 ± 0.019	0.27	0.016	0.677 ± 0.261	0.59	0.274
TFT + EWC, F0	0.472 ± 0.020	0.25	0.019	0.621 ± 0.257	0.62	0.176
TFT joint†	0.455 ± 0.017	0.28	–	0.427 ± 0.049	0.74	–

A. Continual stream (P1)

Table III gives the final average error, skill and forgetting.

S1. TFT + EWC reaches 0.462 pp (skill 0.27), level with gradient boosting (0.466 pp), TFT fine-tuning (0.467 pp) and MLP replay (0.475 pp); a ridge regression on the planned usage alone is slightly better (0.450 pp). Forgetting is small for every method (0.016–0.040 pp). Persistence and linear extrapolation degrade quickly with the horizon (0.83 and 0.90 pp at h=3), while the learned models stay between 0.42 and 0.49 pp. Comparisons with TFT + EWC whose significance changed in a re-run differ by only 0.005–0.014 pp, an order of magnitude below the label shift of the pulse-charge definition (§IV-C); we therefore report S1 method differences as practically equivalent.

S6. When tasks differ in usage, forgetting is substantial (0.18–0.46 pp). EWC lowers the TFT's mean error and mean forgetting relative to fine-tuning (0.677 against 0.883 pp), but which of the two differences is significant changed between runs (§IV-C), so we do not claim a significant benefit. TFT + EWC is not the most accurate method: MLP fine-tuning (0.531 pp) and MLP + EWC (0.540 pp) are significantly better in both runs, the joint upper bounds (0.414 and 0.427 pp) put the TFT's continual-learning cost at about 0.25 pp, and linear extrapolation of the recent fade rate (0.343 pp) and gradient boosting (0.372 pp) are the best forecasters. The TFT varies strongly with task order (s.d. 0.26 pp with EWC, 0.54 pp with fine-tuning), and the diagnostic features do not help it (0.621 pp without them).

B. Unseen cell or usage condition (P2)

Gradient boosting is best on both datasets (Table IV). On S1 the MLP, the TFT and the usage-only ridge are close (0.285–0.301 pp); on S6 the 100% depth-of-discharge condition is the hardest for every method (1.8–2.4 pp). The reuse class of the forecast SOH agrees with the measured class for 95–97% of S6 samples with the learned models (macro-F1 0.83–0.88), against 85% (0.64) for persistence. With six or seven held-out groups, no difference with the TFT survives Holm correction.

TABLE IV

PROTOCOL P2 (UNSEEN BATTERY IN S1, UNSEEN USAGE CONDITION IN S6): MEAN TEST RMSE (PP) OVER HELD-OUT GROUPS, RANGE OVER GROUPS, AND FOR S6 REUSE-CLASS ACCURACY / MACRO-F1 OF THE FORECAST SOH.

Data	Method	RMSE	Range	Class acc./F1
S1	Persistence	0.545	0.436–0.688	–
S1	Linear extrapolation	0.511	0.284–0.725	–
S1	Ridge, usage only	0.301	0.172–0.409	–
S1	Ridge, F0	0.418	0.158–0.779	–
S1	Ridge, F1	0.320	0.161–0.477	–
S1	Gradient boosting, F1	0.219	0.091–0.299	–
S1	MLP	0.285	0.149–0.389	–
S1	TFT	0.292	0.199–0.377	–
S6	Persistence	2.817	2.025–4.327	0.85 / 0.64
S6	Linear extrapolation	1.372	0.856–1.669	0.94 / 0.75
S6	Ridge, usage only	1.177	0.870–2.288	0.95 / 0.83
S6	Ridge, F0	0.859	0.598–1.836	0.96 / 0.86
S6	Ridge, F1	1.137	0.588–2.188	0.96 / 0.85
S6	Gradient boosting, F1	0.710	0.395–1.839	0.97 / 0.88
S6	MLP	0.821	0.476–2.225	0.96 / 0.87
S6	TFT	0.868	0.551–2.390	0.96 / 0.88

C. Reproducibility of the comparisons

P1 and P2 were re-run in a pinned environment on data rebuilt from the raw archives and verified against recorded checksums. Every baseline, both joint upper bounds, every P2 result and the ranking of methods reproduced; the neural models moved by at most 0.059 pp (TFT + EWC on S6: 0.677 to 0.704 pp). On S1, five Holm-corrected comparisons with TFT + EWC crossed $p = 0.05$. On S6, validation selected EWC $\lambda = 1000$ instead of 100, a 0.008 pp tie over three tuning runs; the error advantage of TFT + EWC over TFT fine-tuning then moved from $p_{\mathrm{Holm}} = 0.005$ to 0.060, while its reduction of forgetting moved from 0.050 to 0.008. These findings motivated the ten tuning orders, the one-standard-error rule and the replication rule used for P3.

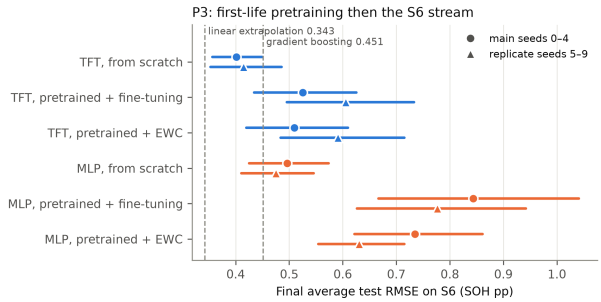


Fig. 3. P3: final average test RMSE on S6, mean and 95% bootstrap CI over 50 runs per seed set. Dashed: linear extrapolation and gradient boosting.

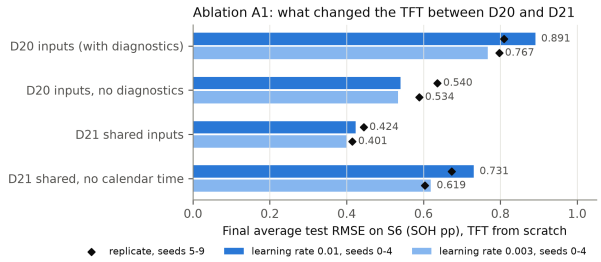


Fig. 4. Post hoc ablation of the TFT trained from scratch on S6: final average test RMSE for four input sets and two learning rates. Bars: mean of 50 runs on seeds 0–4; diamonds: replicate seeds 5–9.

D. First-life pretraining (P3)

The pool provides 36,711 samples, against 2,409 in S6. Validation selected EWC $\lambda = 1000$ for the MLP but only 0.01 for the pretrained TFT. Under the replication rule (Fig. 3):

- EWC and fine-tuning are practically equivalent for the pretrained TFT in final error, forgetting and backward transfer (differences of 0.015–0.017 pp in both seed sets);
- pretrained TFT + EWC beats pretrained MLP fine-tuning (by 0.334 and 0.185 pp), also in forgetting and backward transfer;
- both backbones are better from scratch on average (TFT 0.401/0.415 pp against 0.509/0.591 pp pretrained with EWC), significant for the TFT only in the replicate set;
- linear extrapolation (0.343 pp) remains the most accurate method.

The S6 stream does not degrade the pool test error (TFT + EWC: 3.280 to 3.148 pp), but on that split the pretrained models are less accurate than linear extrapolation (3.001 pp): with the shared inputs, pretraining does not learn a first-life model worth transferring. Pretraining takes 22 s (MLP) and 251 s (TFT) on one CPU thread; EWC adds about 50% to the TFT's stream time without improving accuracy.

Why the TFT improved. A post hoc ablation, repeated on the replicate seeds (Fig. 4), explains why the TFT trained from scratch improves from 0.891 pp with the earlier inputs to 0.401 pp with the shared inputs: dropping the 17 diagnostic features lowers the error by 0.17–0.35 pp and adding the calendar time between checks by 0.19–0.31 pp, both replicated at both learning rates ($p_{\text{Holm}} \leq 0.002$). The learning rate has no significant effect in either seed set.

What the TFT uses. Its top-weighted input changes between seeds, inputs constant within S6 receive weight, and its attention over the four previous origins is nearly uniform (0.24–0.26 per lag): the weights are not a reliable measure of importance here, and the TFT makes little use of its history.

E. Learned residual on linear extrapolation

Table V and Fig. 5 give the results of the residual form (6). The criterion fixed in advance for a positive result, residual TFT + EWC beating linear extrapolation, was *not* met: the two are practically equivalent for horizons 1–3 (differences of -0.004 and -0.029 pp), and for horizons 1–6 extrapolation is better on average ($+0.076$ and $+0.049$ pp, not replicated). Per horizon, extrapolation is better at $h = 1$ and 2 (replicated), and the gap closes without reversing by $h = 5$–6. Learning the residual is, however, what makes the TFT competitive: it improves TFT + EWC over the direct form by 0.17 pp (E1) and 0.24–0.30 pp (E2), and puts it ahead of gradient boosting, all replicated; it is the most accurate of the TFT and MLP variants. With horizons 1–6, where the residual TFT forgets more, EWC lowers its forgetting from 0.28/0.22 to 0.09 pp and its backward transfer from 0.18/0.14 to 0.03 pp, replicated; its effect on the final error is not. For horizons 1–3, EWC and fine-tuning are practically equivalent.

TABLE V
LEARNED RESIDUAL, SHIFT-ADAPTIVE EWC AND LITERATURE BASELINES ON S6 (STREAM FROM SCRATCH): FINAL AVERAGE TEST RMSE (PP), MEAN OVER 50 RUNS, MAIN / REPLICATE SEED SETS, AND FORGETTING (PP) FOR HORIZONS 1–6. BELOW THE LINE: SINGLE FITS ON ALL TASKS (NOT CONTINUAL) AND EXTRAPOLATION.

Method	E1: h=1–3	E2: h=1–6	Forgetting, E2
TFT + EWC, residual	**0.339 / 0.314**	**0.561 / 0.534**	**0.093 / 0.086**
RC-TFT (residual, SA-EWC)	0.343 / 0.321	0.548 / 0.564	0.066 / 0.084
TFT fine-tuning, residual	0.347 / 0.337	0.673 / 0.612	0.282 / 0.220
TFT + EWC, direct	0.504 / 0.487	0.802 / 0.835	0.135 / 0.166
MLP + EWC, residual	0.374 / 0.366	0.659 / 0.694	0.205 / 0.225
BiLSTM + EWC, residual	0.328 / 0.370	0.669 / 0.765	0.229 / 0.318
BiLSTM + EWC, direct	0.323 / 0.323	–	–
Transformer + EWC, residual	0.360 / 0.366	0.609 / 0.705	0.129 / 0.198
TCN + EWC, residual	0.466 / 0.408	0.812 / 0.756	0.345 / 0.268
Linear extrapolation	0.343	0.485	–
Gradient boosting	0.451	0.898	–
SVR, residual	0.429	0.616	–
Random forest, direct	0.487	0.946	–
GPR, direct	1.374	2.073	–

F. Shift-adaptive EWC and literature baselines

Table V also gives the second pre-registered experiment. The criterion for SA-EWC was *not* met: RC-TFT and the fixed-λ residual TFT + EWC are practically equivalent in final error and in forgetting for both horizon sets (differences of

Ageing-cycle inputs. Adding the 21 Spark-extracted inputs (Section III-B) to the residual TFT + EWC, as specified before the run, makes it worse: 0.480/0.501 against 0.339/0.314 pp, and worse than linear extrapolation, both replicated; gradient boosting also worsens (0.495 against 0.451 pp).

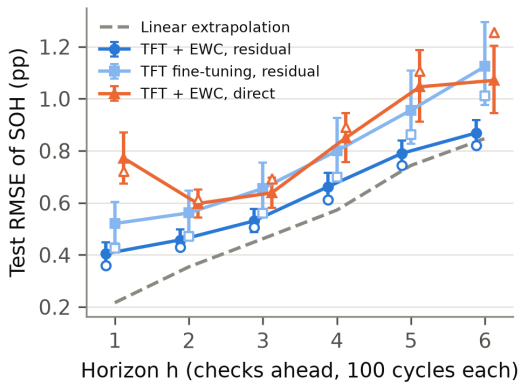


Fig. 5. E2: test RMSE by horizon after the last stream step on S6. Filled markers: mean and 95% bootstrap CI over 50 runs (seeds 0–4); open markers: replicate seeds 5–9. Dashed: linear extrapolation.

at most 0.03 pp), although the shift weights did vary (0.05–1.14). As fixed in advance, the simpler fixed-λ model therefore remains the best-supported variant. Like it, RC-TFT is level with extrapolation for horizons 1–3 and not better for 1–6. The residual TFT is not uniquely accurate at short horizons: BiLSTM + EWC (direct and residual) and the residual Transformer + EWC are practically equivalent to RC-TFT for horizons 1–3. For horizons 1–6, RC-TFT beats the residual BiLSTM and TCN (replicated; the Transformer difference is not replicated). Every single-fit regressor is worse than RC-TFT, replicated, except the residual SVR for horizons 1–6.

G. Shrinkage and intervals on the stream

Shrinking the correction per horizon makes the residual TFT + EWC practically equivalent to extrapolation at every horizon (0.318/0.308 against 0.343 pp for horizons 1–3, 0.498/0.493 against 0.485 pp for 1–6), removing its replicated loss at $h = 1$–2; the correction is turned down most at $h = 1$ (median $\alpha_1 = 0.28$ for 1–6). This criterion for a positive result was again not met: the model matches extrapolation, it does not beat it. Its 90% intervals are narrower than extrapolation's but under-cover (0.71–0.82, against 0.92–0.95), so the interval scores are equal (practically equivalent or not replicated). Pretraining the residual model on the first-life pool makes it worse for horizons 1–3 (by 0.22 pp, replicated) and does not help for 1–6.

H. Residual model on unseen conditions

Table VI gives the fourth experiment. As registered, the shrunk residual TFT + EWC beats extrapolation at origins with a fade rate, by 0.12/0.12 pp for horizons 1–3 and 0.20/0.18 pp for 1–6 (replicated): the first comparison in which the proposed model beats extrapolation. At the other origins extrapolation is persistence and loses by more than 2 pp. Gradient boosting, however, is more accurate than the TFT by 0.39/0.41 pp and 0.57/0.60 pp (replicated), on six of the seven conditions; only on the 100% depth-of-discharge condition is the TFT better. The same residual TFT trained jointly on the six conditions

TABLE VI

UNSEEN S6 USAGE CONDITION (P2, SEVEN HELD-OUT CONDITIONS): TEST RMSE (PP), MEAN OVER FOLDS, ON ALL ORIGINS AND ON ORIGINS WITH A FADE RATE (H^+). TFT ROWS: MEAN OVER 50 RUNS, MAIN / REPLICATE SEED SETS; OTHER ROWS: SINGLE DETERMINISTIC FITS.

Method	E1: h=1–3 all	E1: h=1–3 H^+	E2: h=1–6 all	E2: h=1–6 H^+
TFT + EWC, residual, shrunk	0.92 / 0.94	0.74 / 0.74	1.62 / 1.65	1.42 / 1.44
TFT + EWC, residual, raw	0.88 / 0.90	0.79 / 0.80	1.59 / 1.62	1.46 / 1.49
TFT, residual, joint	0.57 / 0.58	0.54 / 0.55	1.21 / 1.18	1.22 / 1.19
Gradient boosting	**0.54**	**0.51**	**1.05**	**1.05**
Ridge	0.77	0.69	1.30	1.26
Linear extrapolation	1.37	0.86	2.30	1.62
Persistence	2.82	2.66	4.90	4.73

TABLE VII

90% INTERVALS ON UNSEEN S6 CONDITIONS: COVERAGE POOLED OVER THE SEVEN HELD-OUT CONDITIONS, ITS MINIMUM OVER CONDITIONS, AND THE MEAN INTERVAL SCORE (IS, PP). CALIBRATION ROWS: VALIDATION ROWS (VAL) OR ALL ORIGINS OF THE SIX TRAINING CONDITIONS, WITH LEAVE-ONE-CONDITION-OUT FITS (LOCO) FOR GRADIENT BOOSTING (GBM); M: HALF-WIDTHS PER HORIZON AND STRATUM. TFT (SHRUNK RESIDUAL): RANGE OVER THE TWO SEED SETS.

Forecaster, calibration	E1: h=1–3 cov.	min	IS	E2: h=1–6 cov.	min	IS
TFT + EWC, val	0.66	0.38	6.1–6.3	0.67–0.68	0.40–0.41	10.0–10.3
Extrap., val	0.74	0.40	8.5	0.76	0.44	13.5
Extrap., all	0.90	0.74	7.1	0.89	0.71	11.8
Extrap., all, M	0.90	0.71	4.8	0.90	0.67	8.5
GBM, LOCO	0.90	0.48	**3.2**	0.90	0.38	**6.7**
GBM, LOCO, M	0.90	0.51	3.1	0.89	0.39	6.6

beats its continual form by 0.32–0.44 pp (replicated) and varies four to eight times less across runs. Shrinkage and the raw correction are practically equivalent here. The learning rate and λ come from P1 tuning streams that used validation rows of all seven conditions, which can only favour the TFT.

Intervals on unseen conditions. Table VII combines the fourth and fifth experiments. Intervals calibrated on the pooled validation rows of the training conditions under-cover on the held-out one, for the TFT and for extrapolation alike. Calibrating extrapolation, which has no fitted parameters, on all origins of the same conditions instead of their mid-life validation rows restores 0.89–0.90: the calibration rows, not only the model, caused the shortfall. With leave-one-condition-out calibration, gradient-boosting intervals reach the registered threshold of 0.85 (0.90 in both horizon sets) with the lowest interval scores, lower than calibrated extrapolation's on six of seven conditions (significant for horizons 1–3, $p = 0.047$, but not for 1–6, $p = 0.30$, so not established). The coverage holds over conditions, not for each: on the 100% depth-of-discharge condition, which lies outside the range of the others, it is 0.48 and 0.38. Per-stratum half-widths raise extrapolation's coverage at origins without a fade rate from 0.01–0.02 to 0.91–0.92 and change little for gradient boosting. With these intervals, 87–92% of held-out forecasts lie within one reuse class, and 99.4–99.7% of those classes are correct.

V. Discussion

Why the proposed model does not win. On the continual stream TFT + EWC never beat linear extrapolation. Fade over one to three checks is smooth, so extrapolating the recent fade rate leaves little structure for a sequence model to add; the groups are few and small (21–27 and 135–225 training samples per task in S1 and S6); and the TFT barely uses its history window. EWC behaves as expected where there is something to protect: forgetting is negligible across the near-identical S1 cells and substantial across the S6 usage conditions. Even there its benefit to the direct TFT did not survive a re-run, and after pretraining validation selected almost no EWC. Its clearest benefit came once the TFT learned only the correction to extrapolation: over six horizons, EWC cut that model's forgetting by about two thirds, replicated. Making the penalty follow the measured shift between groups added nothing, plausibly because with seven tasks and a tuned λ_0, per-pair weights between 0.05 and 1.14 change the protection little compared with the choice of λ itself (an interpretation, not tested). The backbone matters less than the target: at horizons 1–3 a BiLSTM or Transformer with EWC is as accurate as the residual TFT, which pulls ahead only over six horizons. On unseen conditions, where extrapolation is weaker, the residual TFT beats it, but the continual constraint itself becomes the main cost: trained jointly, the same model is 0.32–0.44 pp more accurate and close to gradient boosting. A model that is never shown the new condition gains nothing from adapting and still pays for the order of the stream. With battery datasets this small, storing every earlier group is cheap, so the constraint that motivates continual learning did not bind here.

What matters. Building on the best simple forecaster mattered most: the learned residual made TFT + EWC one of the most accurate learned models on the S6 stream and level with extrapolation for horizons 1–3. The ablation also points to the inputs rather than the architecture: the diagnostic features hurt the TFT, and the calendar time between checks helped, consistent with fade per cycle rising with that time at 50 °C in all 20 S6 cells. Usage known in advance is a strong predictor, and forecasts are conditional on it. For intervals, the rows used for calibration mattered as much as the model: residuals from mid-life validation origins of conditions seen in training understate the error on a new condition, while residuals from all origins of conditions held out from fitting reproduce it. Dense measurements between checks did not help: with few cells per condition, 21 extra inputs added more variance than information. Pretraining on 369 first-life cells of other chemistries did not transfer to the LFP cells of S6, although retention of the pool was not the problem.

Reproducibility. Several significance results changed in a re-run on different hardware (Section IV-C); small datasets, many low-variance runs and Holm-corrected tests make such fragility likely. We recommend recording the full software environment, tuning with enough runs to separate neighbouring grid values, and requiring that claimed differences replicate on a second seed set and exceed a practically meaningful threshold.

Reuse and positioning. The reuse-class mapping is informative only when the forecast horizon crosses class boundaries: S1 stays in C0 and the S6 stream's test periods lie in C3, while on unseen S6 conditions the classes agree for 95–97% of samples with the learned models. Intervals make the mapping usable for decisions: a forecast whose interval lies within one class (87–92% of samples) was almost always right, and the others are flagged as uncertain rather than assigned. For a battery population unlike any seen before, such as the 100% depth-of-discharge condition here, neither the forecasts nor the intervals can be trusted, and pooled coverage does not reveal it. The studies of Section II estimate the current SOH or SOC on other cells and with other SOH definitions, so their errors are not directly comparable with ours.

Limitations. The task was changed to forecasting after the first dataset was inspected; the second-life datasets are small; S6 is aged at 50 °C and its "second life" continues the same protocol; SOH is relative to each cell's first valid measurement; forecasts are conditional on planned usage; and the ablation and every later experiment were decided after the main results (all except the ablation were specified and committed before they were run). The single-fit regressors see all tasks at once, and the direct literature backbones were run for horizons 1–3 only. The TFT hyperparameters used on unseen conditions were tuned with validation rows of all seven S6 conditions, and the unseen-condition results rest on seven conditions from one laboratory, with the conditions fixed and only training randomness varied. Leave-one-condition-out intervals were run for gradient boosting and extrapolation only.

VI. Conclusion

We built a leakage-controlled benchmark for forecasting the SOH of second-life lithium-ion cells from five public datasets and evaluated a Temporal Fusion Transformer with Elastic Weight Consolidation as a continual learner, with every follow-up experiment pre-registered and every claimed difference replicated on a second seed set. Learned forecasters beat persistence clearly, and forgetting appears whenever usage conditions differ. Learning a shrunk correction to linear extrapolation made TFT + EWC as accurate as extrapolation on the continual stream, where no model beat it, and more accurate than extrapolation on unseen usage conditions. There, however, gradient boosting fitted on all conditions and the same TFT trained jointly were clearly better: at this data scale, the cost of learning continually outweighed its benefit. Shift-adaptive EWC, first-life pretraining, literature backbones and dense ageing-cycle inputs did not help. Conformal intervals held on unseen conditions only when calibrated on conditions held out from fitting, and not for a condition outside the training range; with them, 87–92% of forecasts fell within one reuse class, over 99% of those correctly, and the rest were flagged as uncertain. Future work should test continual learning where earlier data cannot be stored and populations are larger and

more varied, and should calibrate intervals across laboratories and chemistries.

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